

Linear Algebra for Computing

Vectors, Matrices, Transformations and Vector Spaces

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What is a Vector?

Definition:

A vector is a mathematical quantity that has both **magnitude** and **direction**.

A vector can be represented using components.

For example,

$$\mathbf{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

means that the vector moves:

- 3 units in the x direction
- 2 units in the y direction

Example:

If a person moves 3 m east and 2 m north, the displacement can be represented by

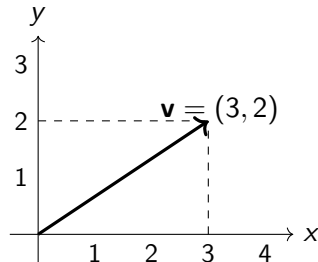
$$\mathbf{v} = (3, 2)$$

Vector Graph

Vector

$$\mathbf{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

The vector starts at the origin $(0,0)$ and ends at $(3,2)$.



Coordinate System

A coordinate system provides a reference for describing the position of points and vectors.

In a two-dimensional coordinate system:

$$P = (x, y)$$

where:

- x represents the horizontal coordinate.
- y represents the vertical coordinate.

Example:

For

$$P = (3, 2)$$

we move 3 units along the x -axis and 2 units along the y -axis.

Vector Addition

Vector addition means adding the corresponding components of two vectors.

If

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

then

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}$$

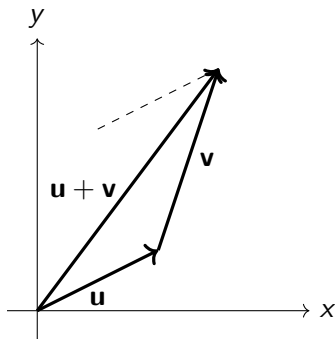
Example:

$$\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Therefore,

$$\begin{bmatrix} 3 \end{bmatrix}$$

Vector Addition Graph



The triangle/parallelogram rule provides a geometric interpretation of vector addition.

Scalar Multiplication

Scalar multiplication means multiplying every component of a vector by the same number.

$$k\mathbf{v} = \begin{bmatrix} kv_1 \\ kv_2 \end{bmatrix}$$

Example:

Let

$$\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

and $k = 3$.

Then

$$3\mathbf{v} = 3 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

The direction remains the same while the magnitude becomes three times larger.

Linear Combination

A linear combination is an expression formed by multiplying vectors by scalars and adding the results.

$$\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2$$

where a and b are scalars.

Example:

Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Then

$$2\mathbf{v}_1 + 3\mathbf{v}_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Therefore, $(2, 3)$ is a linear combination of the two basis vectors.

Span

The **span** of a set of vectors is the collection of all vectors that can be created using their linear combinations.

For example,

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Their span is

$$\text{span}\{\mathbf{e}_1, \mathbf{e}_2\} = \mathbb{R}^2$$

because every vector

$$\begin{bmatrix} x \\ y \end{bmatrix} = x\mathbf{e}_1 + y\mathbf{e}_2$$

can be generated.

Basis Vectors

A basis is a set of vectors that:

- ① spans the space, and
- ② is linearly independent.

For $\mathbb{R}^2$, the standard basis is

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Every vector in $\mathbb{R}^2$ can be written as

$$\mathbf{v} = x\mathbf{e}_1 + y\mathbf{e}_2.$$

Matrix as a Transformation

A matrix can transform a vector from one position to another.

If

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

and

$$\mathbf{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

then

$$A\mathbf{v} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

The matrix stretches the vector in the x direction.

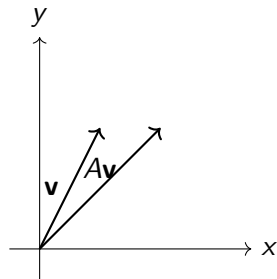
Matrix Transformation Graph

$$\mathbf{v} = (1, 2)$$

After transformation:

$$A\mathbf{v} = (2, 2)$$

The x coordinate is doubled.



Matrix Multiplication as Composition

Two transformations can be applied one after another.

Suppose

$$A\mathbf{v}$$

is the first transformation and

$$B(A\mathbf{v})$$

is the second transformation.

Therefore,

$$B(A\mathbf{v}) = (BA)\mathbf{v}.$$

Thus, matrix multiplication represents the composition of linear transformations.

$$\boxed{BA}$$

means that A is applied first and B is applied second

Three-Dimensional Linear Transformation

A three-dimensional vector has three components:

$$\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

A 3×3 matrix can transform this vector.

For example,

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

scales the x and y coordinates by 2.

Determinant

The determinant is a scalar value associated with a square matrix.
For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

the determinant is

$$\det(A) = ad - bc$$

Example:

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$$

Therefore,

$$\det(A) = (2)(4) - (1)(3) = 8 - 3 = 5.$$

Geometric Meaning of Determinant

For a two-dimensional transformation, the absolute value of the determinant represents the factor by which area changes.

If

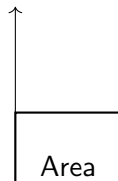
$$|\det(A)| = 5,$$

then an area of 1 becomes an area of 5.

If

$$\det(A) = 0,$$

the transformation collapses the plane into a lower-dimensional space.



Inverse Matrix

The inverse of a matrix A is written as

$$A^{-1}.$$

It satisfies

$$AA^{-1} = A^{-1}A = I.$$

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

The inverse exists only when

$$\det(A) \neq 0$$

Column Space

The column space of a matrix is the set of all linear combinations of its columns.
For

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

the columns are

$$\mathbf{c}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{c}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Therefore,

$$\text{Col}(A) = \mathbb{R}^2.$$

Null Space

The null space contains all vectors $\mathbf{x}$ satisfying

$$A\mathbf{x} = \mathbf{0}.$$

Example:

Let

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}.$$

We solve

$$A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

This gives

$$x + y = 0.$$

Nonsquare Matrices

A nonsquare matrix has different numbers of rows and columns.
For example,

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

is a 3×2 matrix.
It maps

$$\boxed{\mathbb{R}^2 \rightarrow \mathbb{R}^3}$$

because a two-dimensional vector is multiplied by the matrix to produce a three-dimensional vector.

Dot Product

The dot product of two vectors is

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n.$$

For two-dimensional vectors,

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

we have

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2.$$

Example:

$$\mathbf{u} = (2, 3), \quad \mathbf{v} = (4, 1)$$

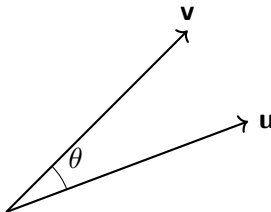
Therefore,

Geometric Meaning of Dot Product

The dot product can also be written as

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}| |\mathbf{v}| \cos \theta$$

where θ is the angle between the vectors.



If

$$\mathbf{u} \cdot \mathbf{v} = 0,$$

then the vectors are perpendicular.

Duality

Duality describes the relationship between vectors and linear functions that act on vectors. A linear functional can be represented as

$$f(\mathbf{x}) = \mathbf{a}^T \mathbf{x}.$$

For

$$\mathbf{a} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix},$$

we get

$$f(\mathbf{x}) = 2x + y.$$

Thus, a vector can define a linear measurement of another vector.

Cross Product

The cross product is defined for vectors in $\mathbb{R}^3$.

For

$$\mathbf{u} = (u_1, u_2, u_3)$$

and

$$\mathbf{v} = (v_1, v_2, v_3),$$

the cross product is

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}.$$

The resulting vector is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$.

Cross Product Example

Let

$$\mathbf{u} = (1, 0, 0)$$

and

$$\mathbf{v} = (0, 1, 0).$$

Then

$$\mathbf{u} \times \mathbf{v} = (0, 0, 1).$$

Therefore,

$$\boxed{\mathbf{u} \times \mathbf{v} = \mathbf{k}}$$

The resulting vector is perpendicular to both original vectors.

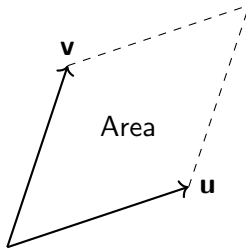
Cross Product and Transformations

The cross product has a geometric meaning related to area.

The magnitude is

$$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}| \sin \theta.$$

This represents the area of the parallelogram formed by $\mathbf{u}$ and $\mathbf{v}$.



Cramer's Rule

Cramer's Rule is a method for solving a system of linear equations using determinants.
Consider

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2.$$

The determinant is

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}.$$

Then

$$x = \frac{D_x}{D}, \quad y = \frac{D_y}{D}.$$

Cramer's Rule Example

Solve:

$$2x + y = 5$$

$$x - y = 1.$$

Main determinant:

$$D = \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -3.$$

For x :

$$D_x = \begin{vmatrix} 5 & 1 \\ 1 & -1 \end{vmatrix} = -6.$$

Therefore,

-6

Change of Basis

A basis provides a coordinate system for representing vectors.

The standard basis of $\mathbb{R}^2$ is

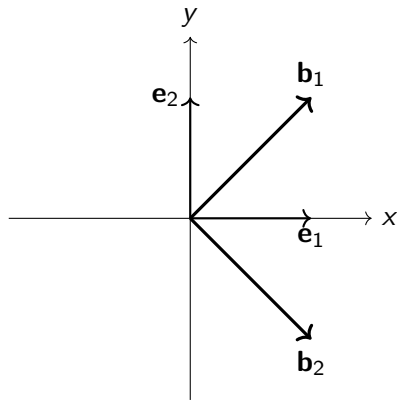
$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Suppose another basis is

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

The same vector can have different coordinates depending on the chosen basis.

Change of Basis Graph



Different bases describe the same vector space using different coordinate directions.

Eigenvectors and Eigenvalues

An eigenvector of a matrix is a non-zero vector whose direction does not change after the transformation.

It satisfies

$$A\mathbf{v} = \lambda\mathbf{v}$$

where:

- $\mathbf{v}$ is the eigenvector.
- λ is the eigenvalue.
- A is the transformation matrix.

The eigenvalue tells us how much the eigenvector is stretched or compressed.

Eigenvalue Example

Let

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

Consider

$$\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Then

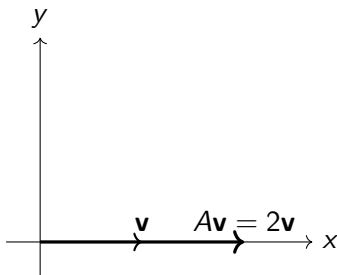
$$A\mathbf{v} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Therefore,

$$\boxed{\lambda = 2}$$

and

Eigenvector Graph



The direction remains unchanged; only the length changes.

Vector Space

A vector space is a collection of objects called vectors that can be added together and multiplied by scalars while satisfying certain rules.

Examples include:

- $\mathbb{R}^2$
- $\mathbb{R}^3$
- Polynomial spaces
- Matrix spaces
- Function spaces

For example,

$$\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}.$$

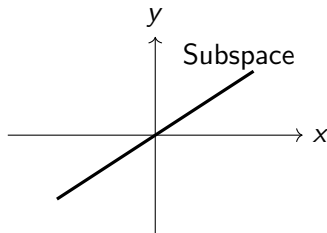
Subspace

A subspace is a subset of a vector space that is itself a vector space.

For a subset W to be a subspace:

- 1 $\mathbf{0} \in W$
- 2 It must be closed under addition.
- 3 It must be closed under scalar multiplication.

For example, a line passing through the origin is a subspace of $\mathbb{R}^2$.



Applications of Linear Algebra

Linear algebra is widely used in computing and data science.

- **Machine Learning**

Vectors represent data and matrices represent transformations.

- **Computer Graphics**

Matrices are used for rotation, scaling, translation and projection.

- **Computer Vision**

Images can be represented using matrices.

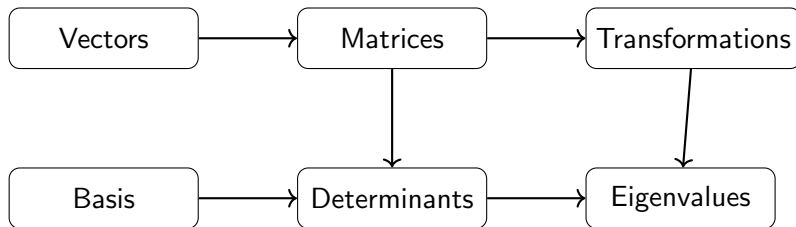
- **Data Science**

Datasets can be represented as matrices.

- **Deep Learning**

Neural network operations involve matrix multiplication.

Big Picture



These concepts are connected and form the foundation of linear algebra used in computing.

Summary

- Vectors represent magnitude and direction.
- Coordinate systems provide a way to represent vectors and points.
- Vector operations include addition and scalar multiplication.
- Linear combinations generate new vectors.
- Span describes the set of vectors that can be generated.
- A basis provides an independent coordinate system.
- Matrices represent linear transformations.
- Determinants describe area and volume scaling.
- Inverse matrices reverse transformations.
- Column space and null space describe important properties of matrices.
- Dot products measure projection and orthogonality.
- Cross products produce perpendicular vectors in $\mathbb{R}^3$.
- Eigenvectors preserve their direction under a transformation.
- Vector spaces provide the mathematical framework for all these concepts.

Thank You